2025/2026

Programming for Data Science

Deadline – April 2nd 2026 @ (23:59)

I. General description

A. Project Goal

The goal of the data science project is to help students **understand the impact of the different choices** made during the data science process (*KDD process*) when training classification models on numeric data.

B. Delivery

Students only need to deliver **a report** describing the results obtained from exploring the dataset and completing the tasks. The report should contain a technical description of the procedures performed on the data, the corresponding results, the decisions made, and possible justifications for those results, according to the template provided.

You can imagine, you are writing a report to be read by your supervisor, not your client, and so the description shall be technical and not from the domain point of view.

The report may be written in Portuguese or English, but it must follow the **template**, provide all the required charts, and not exceed the number of characters allowed per section. Exceeding text will not be considered. Additional charts are allowed and considered.

The report file should be named **`report_X.pdf`** (replacing X with the team number) and submitted through **Fénix** before the deadline stated on the first page.

II. WORK TO DEVELOP

The project consists of performing only **the first iteration of the KDD process** when training a set of models on a single dataset, without considering any additional iterations. In particular, data profiling, data preparation, modeling, and evaluation steps have to be performed.

The goal is not only to describe the best models learnt, but also to understand the impact of the available options on the performance of the trained models.

Students should develop the project in **Python**, using the packages introduced in the classes, namely *matplotlib*, *pandas,* and *scikit-learn*.

The dataset for the task is available on Kaggle, in a private competition created for this purpose.

https://www.kaggle.com/competitions/explicit-song-tracks

If you would like to join this competition, you can use the following invitation link:

https://www.kaggle.com/t/323aa24a71ee4ee28d52ab0000e7effc

Data Extraction

In this first task, the goal is to retrieve the data from the provided source files and store them in appropriate data structures. Then these data structures should be joined into a single table, with *n* rows and *m* columns, where *n* is the number of samples and *m* is the number of features.

Non-numeric variables must be discarded, since scikit-learn doesn't handle them. Processing them is out of the scope of the project, but students may process variables `album_type`, `release_date` and `artist_genres`, if they want. The code used to process them shall be included in the report.

Data Profiling

For this task, data should be characterized along three perspectives: *dimensionality*, *distribution,* and *sparsity*.

Remember that data profiling is used to best understand the data and to identify the required transformations to apply to the original data in the following step. These transformations aim to improve the performance of classification techniques used during the modeling phase.

Students should perform a statistical analysis of the dataset in advance and summarize relevant implications in the report, including the underlying distributions and hypothesize feature dependencies.

Data Preparation

At this stage, the data shall be transformed to address the problems identified in the previous task. For this purpose, students are asked to apply preparation techniques in a predefined order (shown in Figure 1) to minimize the number of datasets to analyze.

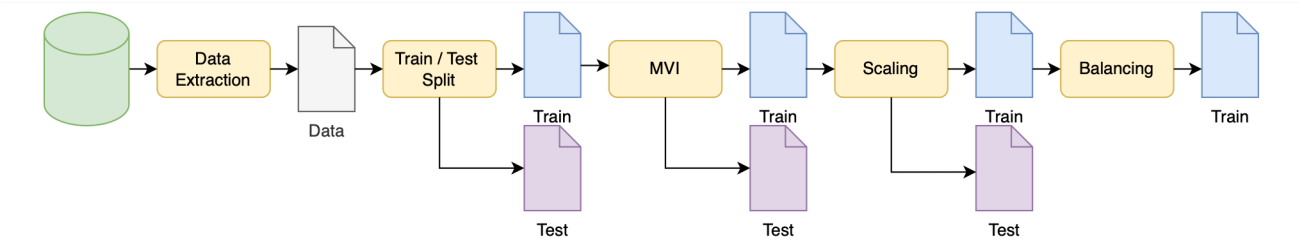

Figure 1 Data preparation methodology for the classification task

The proposal is to process each alternative transformation, then assess the impact of the resulting datasets on training a Naïve Bayes classification model, by estimating its performance.

In this manner, for each preparation step, students must apply **two different preparation techniques** to a single preparation task to obtain different prepared datasets. With each one of these datasets, you train a Naïve Bayes model. Then you compare the results across the different datasets, identify the one that yields the best results, and proceed with the chosen one to the next preparation step.

After training the Naïve Bayes models, we choose the preparation technique that presents the best improvement when compared with the previous dataset. In this manner, after the training, we may face 3 possibilities:

- None of the alternative preparation techniques applied improves the results: so, we should **keep the previous dataset** and proceed to the next step.
- One of the alternatives gives better models, so we **proceed with the dataset** resulting from this transformation.
- Improvements are residual: it is our choice to continue with the previous dataset or with one of the new ones, if it theoretically should yield greater improvements for some of the modeling techniques to be applied.

Remember, you should only apply the technique if the data requires it. For example, if a dataset has no missing values, there is no need to perform missing values imputation. Although, this has to be mentioned and justified in the report.

Some additional remarks:

- It is not possible to train models on missing values with `sklearn`; therefore, the original dataset must be replaced with one of the prepared datasets to proceed to the next step.
- Theoretically, scaling shouldn't change the results for Naïve Bayes; choosing to apply it shall be justified adequately.
- Balancing must be applied only to the training dataset; consequently, data partitioning must be performed beforehand.

Modeling

During the modeling step, students are asked to train a set of classification models to learn the concept identified by the target variable for the dataset. In particular, students have to apply several machine learning methods and corresponding training algorithms, namely: ***Naïve Bayes***, ***Logistic Regression***, ***KNN***, ***Decision Trees,*** and ***Random Forests***.

Again, the goal is to study the impact of the available options. Here, it will be important to study the different parameterizations for each training algorithm.

The use of automatic optimizations offered by *autoML* frameworks is strongly discouraged, since they find the best parameters but do not provide intermediate results, making it impossible to perform the impact analysis required.

The training data shall be the same for all the training methods, corresponding to the result of the preparation step – the dataset that led to the best performance of Naïve Bayes.

Evaluation

Evaluation of the obtained models should be conducted as usual using confidence measures and evaluation charts. A thorough comparison of the models' adequacy should be presented, considering their behavior with respect to the dataset's properties and their observed performance.

For this purpose, the analysis of each classification technique should be done at three different levels:

- The analysis of the impact of the different parameters on the models' performance.
- The description of the best model found for each classification technique, and its performance.
- The identification of the relevant variables, when possible.

Critical Analysis

After identifying the best models learnt with the different ML methods, a critical analysis shall be presented. Students shall compare the best models for each method in terms of content and performance. This analysis may incorporate an individual explanation for each model found, but also **a cross-analysis** of the different results.

III. EVALUATION CRITERIA

The project will be evaluated according to the following decomposition:

EVALUATION CRITERIA		
Data extraction	Reading Files	5%
	Joining Tables	5%
	*(**Optional**) non-numeric variables*	*5%*
Data profiling	Dimensionality	5%
	Distribution	5%
	Sparsity	5%
Data preparation	Missing values imputation	5%
	Scaling	5%
	Balancing	5%
Modeling	Naïve Bayes	5%
	Logistic Regression	5%
	KNN	5%
	Decision Trees	7.5%
	Random Forests	7.5%
Critical analysis		30%

Good Work ! ! !